

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\`  
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ada47fd932d02d5cd.jsonl`  
- SHA-256: `dc118af355c3f1a1b663e12217686ce8ce2439fb2dc3dae7ecd22e31127b9e24`  
- Source modified: `2026-05-31T04:26:22+00:00`  
- Imported at: `2026-07-05T16:48:21+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:25:09.029Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"PP-StructureV3's lightweight configuration runs with roughly 7.4 GB peak VRAM (7456 MB) at ~0.61s average latency per page on an A100, making the full layout+table pipeline feasible on consumer 8-12 GB NVIDIA GPUs."

****Source:**** <https://github.com/PaddlePaddle/PaddleOCR/blob/main/docs/version3.x/algorithm/PP-StructureV3/PP-StructureV3.en.md> (primary)

****Supporting quote:**** "PP_StructureV3-lightweight on Intel 8350C + A100: Peak VRAM 7456 MB, Avg VRAM 7131.95 MB, Avg latency 0.61s"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:25:11.594Z)

I'll verify this claim about PP-StructureV3's lightweight configuration VRAM and latency. Let me start by examining the primary source.

3. assistant (2026-05-31T04:25:14.853Z)

I have the tools loaded. Let me fetch the primary source and search for corroborating/contradicting evidence in parallel.

4. user (2026-05-31T04:25:21.826Z)

PP-StructureV3 Lightweight Configuration Benchmarks

The document contains a comprehensive benchmark table for "PP_StructureV3-lightweight" configurations. Here are the key findings:

Server GPU Results (A100)

For Intel 8350C + A100 hardware:

- **Peak VRAM**: "7456.00" MB
- **Average VRAM**: "7131.95" MB
- **Average latency**: "0.61" seconds per page

Consumer GPU Results (T4)

For Intel 6271C + T4:

- **Peak VRAM**: "8218.00" MB
- **Average VRAM**: "7902.04" MB
- **Average latency**: "1.13" seconds per page

The table header specifies these benchmarks used "PaddleX @ develop" with "CUDA 11.8, cuDNN 8.9" and tested on "280 images including tables, seals, formulas, and charts."

Key distinction: The A100 results represent enterprise server-grade hardware, while the T4 results reflect a more accessible consumer/cloud GPU option with comparable memory efficiency but notably higher latency.

5. user (2026-05-31T04:25:24.341Z)

Web search results for query: "PP-StructureV3 lightweight VRAM 7456 MB latency benchmark A100"

Links: [{"title": "PP-StructureV3 Introduction - PaddleOCR Documentation", "url": "https://paddlepaddle.github.io/PaddleOCR/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html"}, {"title": "NVIDIA A100 CUDA Performance Guide: Specs, Benchmarks & Optimization | RightNow AI | RightNow AI", "url": "https://www.rightnowai.co/guides/gpu-comparison/a100"}, {"title": "NVIDIA A100 TENSOR CORE GPU | DATA SHEET | Jun21 | 1", "url": "https://www.nvidia.com/content/dam/en-zz/Solutions/Data-Center/a100/pdf/nvidia-a100-datasheet-us-nvidia-1758950-r4-web.pdf"}, {"title": "A100 GPU's Offer Power, Performance, & Efficient Scalability", "url": "https://www.nvidia.com/en-us/data-center/a100/"}, {"title": "A100 40GB vLLM Benchmark: Cheap GPU for Gemma3 Hosting", "url": "https://www.databasemart.com/blog/vllm-gpu-benchmark-a100-40gb"}, {"title": "NVIDIA A100 TENSOR CORE GPU | DATA SHEET | 1", "url": "https://www.nvidia.com/content/dam/en-zz/Solutions/Data-Center/a100/pdf/nvidia-a100-datasheet-nvidia-us-2188504-web.pdf"}, {"title": "NVIDIA A100 vs V100: Full Specs, Benchmarks, and GPU Comparison for AI Workloads | Spheron Blog", "url": "https://www.spheron.network/blog/nvidia-a100-vs-v100/"}, {"title": "Nvidia H100 GPU: Specs, VRAM, Price, and AI Performance", "url": "https://www.runpod.io/articles/guides/nvidia-h100"}]

Based on the search results, I found information about PP-StructureV3 and A100 benchmarking:

Local inference was tested with both V100 and A100 GPU, evaluating the performance of PP-StructureV3 under 6 different configurations. The test data consists of 15 PDF files, totaling 925 pages, including elements such as tables, formulas, seals, and charts.

Additionally, the serving inference test is based on the NVIDIA A100 + Intel Xeon Platinum 8350C environment, with test data consisting of 1500 images, including tables, formulas, seals, charts, and other elements.

However, the search results do not contain specific details about the 7456 MB VRAM or specific latency benchmark numbers for the lightweight configuration you're asking about. For more detailed benchmark specifications, you may want to check the official PaddleOCR documentation or benchmark pages directly, as the results reference configuration details that may be available in tables or detailed sections of their documentation.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:25:28.957Z)

The WebFetch confirms the exact numbers from the primary source. The supporting quote is accurate. However, the critical question for the claim is the inference: does 7.4 GB peak VRAM on an A100 mean the pipeline is "feasible on consumer 8-12 GB NVIDIA GPUs"?

Notably, the T4 (a consumer/datacenter GPU with 16GB) shows the SAME lightweight config using 8218 MB peak VRAM ? already exceeding the lower bound of "8-12 GB". Let me dig into this VRAM discrepancy and whether 8GB consumer cards are actually viable.

7. user (2026-05-31T04:25:32.770Z)

REDIRECT DETECTED: The URL redirects to a different host.

Original URL: <https://paddlepaddle.github.io/PaddleOCR/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html>
Redirect URL: <http://www.paddleocr.ai/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html>
Status: 301 Moved Permanently

To complete your request, I need to fetch content from the redirected URL. Please use WebFetch again with these parameters:

- url: "http://www.paddleocr.ai/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html"
- prompt: "I need the FULL benchmark table for PP-StructureV3 configurations. List EVERY row: each configuration name (lightweight, standard, high-performance, etc.), the GPU used (A100, V100, T4), peak VRAM in MB, average VRAM in MB, and average latency in seconds. I especially need: (1) the lightweight config peak VRAM on EVERY GPU listed, not just A100. (2) Whether the lightweight config uses a GPU or CPU. (3) Any footnotes about batch size, precision (FP16/FP32), or whether these are GPU-only memory or total. (4) Does the documentation say anywhere that this fits on 8GB consumer GPUs? Quote exact numbers."

8. user (2026-05-31T04:25:47.354Z)

Web search results for query: "PP-StructureV3 GPU memory requirement consumer GPU RTX 3060 8GB out of memory"

Links: [{"title": "GeForce RTX 3060 Family | NVIDIA", "url": "https://www.nvidia.com/en-us/geforce/graphics-cards/30-series/rtx-3060-3060ti/"}, {"title": "Nvidia RTX 3060 - GPU benchmark - UserBenchmark", "url": "https://gpu.userbenchmark.com/Nvidia-RTX-3060/Rating/4105"}, {"title": "NVIDIA GeForce RTX 3060 8GB GPU upgraded to 12GB memory gets 22% boost in Unigine 1080p Extreme benchmark - VideoCardz.com", "url": "https://videocardz.com/newz/nvidia-geforce-rtx-3060-8gb-gpu-upgraded-to-12gb-memory-gets-22-boost-in-unigine-1080p-extreme-benchmark"}, {"title": "NVIDIA GeForce RTX 3060 Desktop GPU - Benchmarks and Specs - Notebookcheck Tech", "url": "https://www.notebookcheck.net/NVIDIA-GeForce-RTX-3060-Desktop-GPU-Benchmarks-and-Specs.579000.0.html"}, {"title": "NVIDIA GeForce RTX 3060 GPU", "url": "https://www.itcreations.com/nvidia-gpu/nvidia-geforce-rtx-3060-gpu"}, {"title": "NVIDIA GeForce RTX 3060 with 8GB memory has been tested, 17% performance difference vs. 12GB model - VideoCardz.com", "url": "https://videocardz.com/newz/nvidia-geforce-rtx-3060-with-8gb-memory-has-been-tested-17-performance-difference-vs-12gb-model"}, {"title": "Modded NVIDIA GeForce RTX 3060 GPU Gets Memory Boost From 8 GB To 12 GB, Delivers 22% Higher Performance", "url": "https://wccftech.com/modded-nvidia-geforce-rtx-3060-gpu-memory-boost-8-gb-to-12-gb-delivers-22-percent-higher-performance/"}, {"title": "GeForce RTX 3060 [in 18 benchmarks]", "url": "https://technical.city/en/video/GeForce-RTX-3060"}, {"title": "GeForce RTX 3060 with 8GB / 128-bit memory bus memory released", "url": "https://www.guru3d.com/news-story/geforce-rtx-3060-with-8gb-128-bit-memory-bus-memory-released.html"}, {"title": "Nvidia GeForce RTX 3060 8GB With 128-Bit Memory Bus Appears | Tom's Hardware", "url": "https://www.tomshardware.com/news/nvidia-geforce-rtx-3060-8gb-with-128-bit-memory-bus"}]

The search results provide information about RTX 3060 GPU specifications, but don't contain specific information about PP-StructureV3 and its GPU memory requirements. Let me search more specifically for PP-StructureV3.

Links: [{"title": "PP-StructureV3 Introduction - PaddleOCR Documentation", "url": "https://www.paddleocr.ai/latest/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html"}, {"title": "Placement of virtual GPU requests in virtual GPU enabled systems using a requested memory requirement of the virtual GPU request", "url": "https://image-pubs.uspto.gov/dirsearch-public/print/downloadPdf/11372683"}, {"title": "PaddleOCR Guide 2026: PP-OCRV3, v4, v5 for Developers", "url": "https://www.tenorshare.com/ocr/paddleocr.html"}, {"title": "(PDF) PaddleOCR 3.0 Technical Report", "url": "https://www.researchgate.net/publication/393511573_PaddleOCR_30_Technical_Report"}, {"title": "PP-StructureV3 - a PaddlePaddle Collection", "url": "https://huggingface.co/collections/PaddlePaddle/pp-structurev3"}, {"title": "PaddleOCR 3.0 Technical Report", "url": "https://arxiv.org/html/2507.05595v1"}]

Let me search for more specific information about PP-StructureV3 model specifications and memory requirements.

Links: [{"title": "PP-StructureV3 Introduction - PaddleOCR Documentation", "url": "https://paddlepaddle.github.io/PaddleOCR/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html"}, {"title": "Githubusercontent", "url": "https://raw.githubusercontent.com/PaddlePaddle/PaddleOCR/main/docs/version3.x/algorithm/PP-StructureV3/PP-StructureV3.en.md"}, {"title": "Free LLM VRAM Calculator: GPU Memory Requirements for Large Language Models - Free Online Tool", "url": "https://simulations4all.com/simulations/llm-vram-calculator"}, {"title": "LLMem: Estimating GPU Memory Usage for Fine-Tuning Pre-Trained LLMs", "url": "https://arxiv.org/pdf/2404.10933"}, {"title": "GPU Memory Prediction for Multimodal Model Training", "url": "https://arxiv.org/pdf/2512.07853"}, {"title": "Estimate GPU Memory (VRAM) - Shayan Geek", "url": "https://shayangeeek.com/en/estimate-gpu-memory-needed-for-ai-model/"}, {"title": "Memory Analysis on the Training Course of DeepSeek Models", "url": "https://arxiv.org/pdf/2502.07846"}, {"title": "Calculating GPU memory for serving LLMs | Continuum Labs", "url": "https://training.continuumlabs.ai/infrastructure/data-and-memory/calculating-gpu-memory-for-serving-llms"}]

Let me search for the actual memory requirement table from PP-StructureV3 documentation, which appears to be referenced in the results.

Links: [{"title": "PP-StructureV3 Introduction - PaddleOCR Documentation", "url": "https://paddlepaddle.github.io/PaddleOCR/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html"}, {"title": "PaddleOCR 3.0 Technical Report", "url": "https://arxiv.org/html/2507.05595v1"}, {"title": "(PDF) PaddleOCR 3.0 Technical Report", "url": "https://www.researchgate.net/publication/393511573_PaddleOCR_30_Technical_Report"}, {"title": "PP-StructureV3 - Document Analysis & Table OCR - a mindchain Collection", "url": "https://huggingface.co/collections/mindchain/pp-structurev3-document-analysis-and-table-ocr"}, {"title": "PP-StructureV3 Introduction - PaddleOCR Documentation", "url": "http://www.paddleocr.ai/main/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html"}]

Based on my search results, I found information about PP-StructureV3 and GPU requirements, though the specific memory table appears to be embedded in the official documentation. Here's what I found:

Summary of Search Results

PP-StructureV3 is a SOTA document parsing solution on OmniDocBench, supporting the conversion of PDFs and document images to Markdown and JSON.

Regarding GPU memory requirements and your RTX 3060 8GB concern:

By default, the largest models for each module are used. Section 3.3 demonstrates how different model selections affect GPU memory consumption and inference speed. You can choose an appropriate model based on your device capabilities and the complexity of your samples.

The key point is that while PP-StructureV3 is recommended to run on GPU for optimal performance, it also supports CPU inference. Thanks to a variety of configuration options and sufficient optimization for lightweight models, users can refer to section 3.3 to select lightweight configurations for CPU-only environments.

If you're experiencing out-of-memory errors with an RTX 3060 8GB, the solution is to select a lightweight model configuration rather than the default largest models. The official PaddleOCR documentation (Section 3.3) contains detailed information about memory consumption for different model variants that would allow you to find a configuration suitable for your 8GB GPU.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:25:49.830Z)

Let me fetch the redirected URL to get the full benchmark table, and check the raw markdown for the complete details.

10. user (2026-05-31T04:25:59.466Z)

PP-StructureV3 GPU Memory Benchmark Analysis

Complete Benchmark Table (Section 3.3)

Based on the pipeline benchmark data provided, here is the full configuration breakdown:

Configuration	Hardware	Peak VRAM (MB)	Avg VRAM (MB)	Avg Latency (s)
PP_StructureV3-default	A100	37,370	34,165.68	1.38
PP_StructureV3-default	V100	26,824	24,581.61	2.38
PP_StructureV3-default	H20	35,132	32,077.12	1.36
PP_StructureV3-default	A10	18,536	18,353.93	1.74
PP_StructureV3-default	T4	13,970	12,668.58	3.70
PP_StructureV3-lightweight	A100	7,456	7,131.95	0.61
PP_StructureV3-lightweight	V100	8,450	8,083.30	1.07
PP_StructureV3-lightweight	H20	7,524	7,167.52	0.46
PP_StructureV3-lightweight	A10	7,248	6,927.58	0.70
PP_StructureV3-lightweight	T4	8,218	7,902.04	1.13

Critical Findings

(1) Lightweight Config on A100 vs T4:
- **A100:** Peak 7,456 MB, Average 7,131.95 MB
- **T4:** Peak 8,218 MB, Average 7,902.04 MB

(2) GPU Pipeline Status:
The lightweight configuration explicitly runs as a GPU pipeline. The table notes indicate GPU utilization percentages (52-89% for lightweight across devices), confirming GPU execution.

(3) Batch Size & Precision:
The document does not specify batch size or precision (FP16/FP32) for these benchmarks.

(4) Consumer GPU Compatibility:
The document **does not state** that any configuration fits within 8GB or 12GB consumer GPU constraints. The smallest VRAM requirement listed is approximately 7.2 GB average (lightweight on A100).

11. user (2026-05-31T04:26:01.500Z)

PP-StructureV3 Benchmark Data Extraction

(1) Complete PP_StructureV3 Configuration Rows with GPU Specs

From the pipeline benchmark table (local inference):

Configuration	GPU	Avg Latency (s)	Peak VRAM (MB)	Avg VRAM (MB)
PP_StructureV3-default	A100	1.38	37,370	34,165.68
PP_StructureV3-default	V100	2.38	26,824	24,581.61
PP_StructureV3-default	T4	3.70	13,970	12,668.58
PP_StructureV3-lightweight	A100	0.61	7,456	7,131.95
PP_StructureV3-lightweight	T4	1.13	8,218	7,902.04

(2) Lightweight Config VRAM

- **A100**: "Peak device memory (MB)" = 7,456; "Avg. device memory (MB)" = 7,131.95
- **T4**: Peak = 8,218 MB; Average = 7,902.04 MB

(3) Batch Size and Precision

The document does not specify batch size or precision settings in the benchmark methodology.

(4) Test Type

All pipeline benchmark data presented are from **local inference** testing, as indicated by section header "## Local inference."

(5) Consumer GPU or Memory Suitability Claims

The document contains **no explicit statements** regarding suitability for consumer GPUs (e.g., 8GB/12GB VRAM). It only provides benchmark data on enterprise-grade hardware (A100, V100, H20, A10, T4).

12. assistant (2026-05-31T04:26:12.146Z)

I now have the complete benchmark table. Let me analyze the claim carefully against the evidence.

The claim has three parts:

1. **"~7.4 GB peak VRAM (7456 MB) ... on an A100"** ? VERIFIED. The table shows exactly 7456 MB peak, 7131.95 MB avg, 0.61s latency on A100.
2. **"making the full layout+table pipeline feasible on consumer 8-12 GB NVIDIA GPUs"** ? This is the inferential leap that needs scrutiny.

Key issues with the inference:

- **VRAM varies significantly by GPU.** The SAME lightweight config uses **8218 MB peak** on a T4 and **8450 MB peak** on a V100 ? both already exceeding 8 GB. So on the lower bound of the claimed "8-12 GB" range, the pipeline would NOT fit on an 8 GB card with headroom (you'd be at or over the limit before accounting for OS/display overhead, CUDA context ~0.5-1 GB, and framework reserved memory).

- The 7456 MB figure is the **best-case** GPU (A100), not representative of consumer cards. Consumer 8-12 GB cards (RTX 3060 12GB, RTX 4060, RTX 3070 8GB) are architecturally closer to T4/A10 than A100. The A10 shows 7248 MB and T4 shows 8218 MB ? so the realistic consumer-GPU peak is ~7.2-8.2 GB, which is tight-to-over for an 8 GB card.

- **The documentation itself makes NO claim of 8-12 GB consumer GPU feasibility.** That conclusion is the claim author's extrapolation, not the source's.

Let me verify the consumer-GPU feasibility angle with one more search to see if there's reported real-world OOM behavior or whether the framework's memory pre-allocation changes the picture.

13. user (2026-05-31T04:26:16.989Z)

You've hit your session limit · resets 1:30pm (Asia/Calcutta)

14. user (2026-05-31T04:26:22.308Z)

Web search results for query: "PaddleOCR PP-Structure GPU memory usage RTX 3060 12GB 8GB lightweight model inference"

Links: [{"title": "Usage Tutorial - PaddleOCR Documentation", "url": "https://www.paddleocr.ai/latest/en/version3.x/pipeline_usage/PaddleOCR-VL.html"}, {"title": "PaddleOCR-VL-0.9B - FastDeploy: Large Language Model Deployment", "url": "https://paddlepaddle.github.io/FastDeploy/best_practices/PaddleOCR-VL-0.9B/"}, {"title": "How to use GPU in paddleocr · Issue #10429 · PaddlePaddle/PaddleOCR", "url": "https://github.com/PaddlePaddle/PaddleOCR/issues/10429"}, {"title": "Frequently Asked Questions on Inference and Deployment of PaddleOCR-VL PaddleOCR-VL ???????????? · Issue #16823 · PaddlePaddle/PaddleOCR", "url": "https://github.com/PaddlePaddle/PaddleOCR/issues/16823"}, {"title": "Is it possible to train ppocr_v4 on a 6GB GPU? I am running out of memory · Issue #10783 · PaddlePaddle/PaddleOCR", "url": "https://github.com/PaddlePaddle/PaddleOCR/issues/10783"}, {"title": "Cracking the ?Vertical Text? Curse: Fine-tuning PaddleOCR-VL for Japanese Manga on an RTX 3060 | by Alex Zhang | Medium", "url": "https://medium.com/@alex_paddleocr/cracking-the-vertical-text-curse-fine-tuning-paddleocr-vl-for-japanese-manga-on-an-rtx-3060-abc129bd0939"}, {"title": "GPU memory is not released after prediction · PaddlePaddle/PaddleOCR · Discussion #6977", "url": "https://github.com/PaddlePaddle/PaddleOCR/discussions/6977"}, {"title": "paddleocr · PyPI", "url": "https://pypi.org/project/paddleocr/"}, {"title": "unexpected GPU and Debug warnings · PaddlePaddle/PaddleOCR · Discussion #14362", "url": "https://github.com/PaddlePaddle/PaddleOCR/discussions/14362"}, {"title": "PP-StructureV3 Introduction - PaddleOCR Documentation", "url": "http://www.paddleocr.ai/v3.0.0/en/version3.x/algorithm/PP-StructureV3/PP-StructureV3.html"}]

I found relevant information about PaddleOCR PP-Structure and GPU memory usage. Here's what the search results show:

Key Findings:

****Hardware Support:****

The minimum supported configuration that successfully runs is an RTX 3060 (12 GB), and the lowest supported Compute Capability is 7.0.

****Memory Usage for PP-Structure:****

The search results provide specific memory metrics for PP-StructureV3. On the V100, the peak and average GPU memory would be reduced from 8776.0 MB and 8680.8 MB to 6118.0 MB and 6016.7 MB, respectively; On the A100, the peak and average GPU memory would be reduced from 11716.0 MB and 11453.9 MB to 9850.0 MB and 9593.5 MB, respectively when using mobile OCR models.

****Memory Variability:****

Memory usage and inference speed may vary significantly across different hardware devices, influenced by the total GPU memory and Compute Capability.

****8GB GPU Configuration:****

The search results indicate that while the RTX 3060 12GB is the officially tested minimum, the 8GB variant is not explicitly mentioned as supported. The documentation confirms RTX 3060 with 12GB works, but lighter weight models would be necessary for the 8GB variant.

****Optimization Tips:****

When using mobile OCR models + PP-FormulaNet_plus-M, if you set use_chart_recognition to False and don't load the chart recognition model, the GPU memory would be reduced.

For precise memory requirements on RTX 3060 specifically, the official documentation appears to focus on V100 and A100 GPUs in the detailed memory benchmarks.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

15. assistant (2026-05-31T04:26:22.720Z)

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